

MARK SCHEME for the May/June 2009 question paper
for the guidance of teachers

9709 MATHEMATICS

9709/06

Paper 6, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the May/June 2009 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only – often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

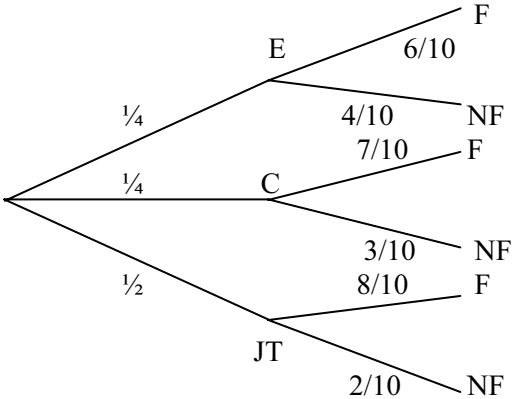
Penalties

MR –1	A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR–2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA –1	This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

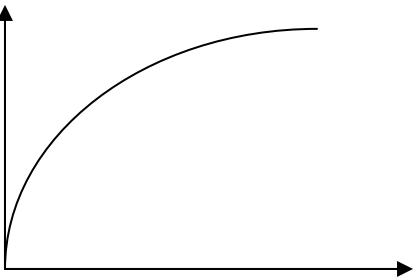
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1	(i) $z = 0.674$ $\frac{1002 - \mu}{8} = 0.674$ $\mu = 997$	B1 M1 A1 [3]	± 0.674 or rounding to, seen, e.g. 0.6743 Standardising and attempting to solve for μ , must use recognisable z-value, no cc, no sq rt, no sq Correct answer rounding to 997																									
	(ii) $P(2) = 3 \times \frac{225}{900} \times \frac{224}{899} \times \frac{675}{898}$ $= 0.140$ OR $\frac{{}^{225}C_2 \times {}^{675}C_1}{{}^{900}C_3}$	M1 A1 [2]	$900 \times 899 \times 898$ or ${}^{900}C_3$ seen in denom Correct answer not 0.141 or 0.14																									
2	(i) $P(X = 2) = 1/4 \times 1/4 + 1/4 = 5/16$ AG OR can use a table <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>1</td><td>2</td><td>2</td><td>4</td><td>4</td></tr><tr><td>2</td><td>3</td><td>2</td><td>5</td><td>4</td></tr><tr><td>3</td><td>4</td><td>2</td><td>6</td><td>4</td></tr><tr><td>4</td><td>5</td><td>2</td><td>7</td><td>4</td></tr></table>		1	2	3	4	1	2	2	4	4	2	3	2	5	4	3	4	2	6	4	4	5	2	7	4	M1 A1 [2]	Considering cases (1, 1) and (2) Correct given answer legitimately obtained (1/16 + 4/16 needs some justification but 1/16 + 1/4 is acceptable)
		1	2	3	4																							
1	2	2	4	4																								
2	3	2	5	4																								
3	4	2	6	4																								
4	5	2	7	4																								
	(ii) $E(X) = \sum xp$ $= 15/4$ (3.75) $\text{Var}(X) = 2^2 \times 5/16 + 3^2 \times 1/16 + 4^2 \times 3/8 + \dots - (15/4)^2$ $= 260/16 - 225/16 = 35/16$ (2.19)	M1 A1 M1 A1 [4]	Using correct formula for $E(X)$, no extra division Correct answer Using a variance formula correctly with mean ² subtracted numerically, no extra division Correct final answer																									
	3	(i) $P(X < 3) = P(0) + P(1) + P(2)$ $= (0.84)^{11} + (0.16)(0.84)^{10} \times {}^{11}C_1 + (0.16)^2(0.84)^9 \times {}^{11}C_2$ $= 0.1469 + 0.30782 + 0.2931$ $= 0.748$	M1 M1 A1 [3]	Binomial term with ${}^{11}C_r p^r (1-p)^{11-r}$ seen Correct expression for $P(0, 1, 2)$ or $P(0, 1, 2, 3)$ Can have wrong p Correct final answer. Normal approx M0 M0 A0																								
	(ii) $\mu = 125 \times 0.64 = 80$ $\sigma^2 = 125 \times 0.64 \times 0.36 = 28.8$ $P(X > 73) = 1 - \Phi\left(\frac{73.5 - 80}{\sqrt{28.8}}\right)$ $= \Phi(1.211)$ $= 0.887$	B1 M1 M1 M1 A1 [5]	80 and 28.8 or 5.37 seen standardising, with or without cc, must have sq rt in denom continuity correction 73.5 or 72.5 only correct region (> 0.5 if mean > 73.5, vv if mean < 73.5) correct answer																									

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4	(i) ${}^{13}C_{10} \times {}^{12}C_9 \times {}^6C_4 \times {}^7C_4$ $= 33033000 (33000000)$	M1 A1 [2]	Expression involving the product of 4 combinations Correct final answer allow 33×10^6 or 3.3×10^7
	(ii) $5! \times 6!$ $= 86400$	B1 M1 A1 [3]	6! or 5! or 4! oe seen no denom a single product involving 6! and either 4! or 5! no denom Correct final answer
	(iii) $4! \times 3! \times 2$ $= 288$	B1 M1 A1 [3]	4! or 3! or 4!/4 seen a single product involving 3! (or 4!/4) and 4! Correct final answer
5	(i) $P(E) = \frac{1}{4}, P(C) = \frac{1}{4}, P(JT) = \frac{1}{2}$	B1 B1 [2]	$\frac{1}{4}, \frac{1}{4},$ and $\frac{1}{2}$ seen oe 3 evaluated probs correctly associated
	(ii) 	M1 A1ft [2]	E, C, JT then F on appropriate shape All probs and labels showing and correct, ft their (i) if $\Sigma p = 1$. If nothing seen in part (i) then give M1 A1ft bod provided their $\Sigma p = 1$ No retrospective marking
	(iii) $P(F) = (\frac{1}{4} \times \frac{6}{10}) + (\frac{1}{4} \times \frac{7}{10}) + (\frac{1}{2} \times \frac{8}{10})$ $= \frac{29}{40} (0.725)$	M1 B1 [2]	Summing 3 appropriate two-factor products provided $\Sigma p = 1$ Correct answer
	(iv) $P(C NF) = \frac{P(C \cap NF)}{P(NF)}$ $= \frac{3/40}{(1 - 29/40)}$ $= 3/11 (0.273)$ OR using ratios $3/(4+3+4)$	B1ft M1 A1 [3]	$1 - 29/40$ seen in denom, ft $1 -$ their (iii) attempt at cond prob with their $C \cap F$ or $C \cap NF$ in numerator correct answer

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6 (i) $a = 494$ $b = 46$	B1 B1 [2]	
(ii) 	B1 B1 M1 A1 [4]	Correct linear scale minimum 0 to 540 and 0 to 60 Labels (cf or people or number of people) and (time, or minutes) and attempt at cf or cf step polygon Attempt to plot points at (10, 210), (20, 344), (30, 422), (40, 494) Correct graph through (0, 0) and (60, 540)
(iii) median is 13.5 to 14.6 min	M1 A1 [2]	Attempt to read from graph at line $y = 270$ or 270.5 Correct answer
(iv) $(5 \times 210 + 15 \times 134 + 25 \times 78 + 35 \times 72 + 50 \times 46) / 540$ $= 9830 / 540$ $= 18.2 \text{ min}$ $(5^2 \times 210 + 15^2 \times 134 + \dots) - 18.2^2$ $\text{sd} = 14.2 \text{ min}$	M1 A1 M1 A1 [4]	Using mid points and frequencies Correct mean Attempt at $\Sigma x^2 f / \Sigma f$ – their mean ² numerically, could use cfs, ucb, but not class widths Correct answer
(v) $18.2 \pm 7.1 = 11.1, 25.3$ $390 - 225$ $= 155 \text{ to } 170 \text{ people}$	M1 A1 [2]	Attempt to read their mean $\pm \frac{1}{2}$ sd from cf graph Correct answer